

Agent-Infrastructure Fit: How AI Agents Are Redefining the Governance of Public Digital Data Infrastructure

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Abstract

As AI agents enter civic analytics, public-sector decision-making, and policy research, the infrastructure quality of government open-data portals becomes a critical governance problem. Existing evaluation frameworks emphasize legal availability, metadata completeness, and human-facing demand, but rarely test whether AI agents can successfully use public data. This paper introduces agent-infrastructure fit: the alignment between task requirements and the affordances a public-data system provides. We argue that agentic use expands open-data governance beyond publication: datasets must also be queryable, joinable, semantically documented, and resilient to machine failure modes. The evidence has three layers. First, Socrata/Tyler portal logs from San Francisco, New York City, and Seattle from 2020 to 2026 provide secondary associational context. High-friction datasets exhibit stronger post-2025 API/download growth relative to page views, most clearly in New York City; the design does not identify a causal AI effect. Second, we develop an Agent-Readable Infrastructure Index (ARI) covering six dimensions: metadata completeness, schema documentation, API queryability, joinability, semantic/reference support, and operational readiness. Third, a controlled benchmark executes 90 civic-data tasks three times under no-metadata, schema-only, and structured-ARI conditions, producing 810 completed runs. Schema access achieves the highest automatic task-success rate (62.2%), compared with structured ARI cards (52.2%) and no metadata (43.0%). Structured ARI cards nevertheless reduce hallucinated fields by 89.5% relative to schema access and reduce wrong joins from 0.15 to 0.02 per run. Automatic validators flag missing reference assets in 59.1% of runs. These results reveal a performance-reliability frontier: concise schemas support completion, while structured semantic constraints reduce critical errors. The paper shifts open-data evaluation from static availability toward relational, task-level agent usability and provides a governance diagnostic for auditing whether public digital infrastructure is ready for AI-mediated analytical work.

Keywords: AI agents; agent-infrastructure fit; open government data; agent-readable infrastructure; digital public infrastructure; data governance

1. Introduction

Open-data programs established publication as the default for many government records. Publication alone does not create public value. People still have to find, interpret, query, combine, and apply the data to a decision. Research on open government data documents persistent breaks in that sequence, including incomplete metadata, weak interoperability, limited technical capacity, and fragmented responsibility (Ansari et al., 2022; Attard et al., 2015; Dawes et al., 2016; Gurstein, 2011; Janssen et al., 2012; Safarov et al., 2017; Wirtz et al., 2018). A dataset may be legally open and available online yet remain difficult to use.

AI coding agents raise the stakes of this familiar problem. They can generate code, call APIs, inspect schemas, and assemble analytic workflows, which may lower the expertise required to reuse civic data. Their performance is uneven across tasks, and small changes in the information supplied to them can shift the quality of their work (Dell'Acqua et al., 2026). An agent asked to analyze a city dataset needs more than a catalog page. It needs a stable resource identifier, a working endpoint, valid field names and syntax, join keys, code dictionaries, boundary files, time coverage, and warnings about ambiguity. When those items are missing or unclear, the agent may invent a field, query the wrong asset, use an invalid join, or omit a required reference table.

We call this relational governance problem agent-infrastructure fit, defined as the degree to which a public-data system's observable affordances match the requirements of a specific agent-mediated task. The construct extends task-technology fit (Goodhue & Thompson, 1995). The focal user is a probabilistic, tool-using agent working with human oversight. The relevant technology includes the portal, API, metadata, schemas, reference assets, operating processes, and governance rules. Performance includes reliability and

traceability as well as completion. A quick answer built on an invented field is a failed civic-data outcome.

The study uses three forms of evidence. The primary analysis is a controlled benchmark of 90 civic-data tasks executed three times under three information conditions, for 810 completed runs. It measures a protocol-defined automatic pass rate, invalid endpoints and queries, hallucinated fields, wrong joins, missing reference assets, elapsed time, and token use. We also construct an Agent-Readable Infrastructure Index (ARI) from six dimensions of dataset readiness. Public Socrata/Tyler portal analytics provide context on changes in access patterns. The portal analysis receives less evidentiary weight because anonymous activity combines people, bots, crawlers, applications, and scripts (Socrata, 2024), and the pre-period diagnostics reject a causal interpretation.

The results show that information breadth and constraint serve different purposes. Complete schema access has the highest automatic pass rate at 62.2%, followed by structured ARI cards at 52.2% and minimal context at 43.0%. The structured ARI card reduces hallucinated fields by 89.5% relative to the schema condition and lowers wrong joins from 0.15 to 0.02 per run. Broad schemas support completion and reference discovery. Compact constraints reduce critical errors but may omit details needed for some tasks. A stricter post-hoc audit requires the model to identify the correct target dataset for schema, API, and substantive-answer tasks. It lowers the minimal-context pass rate to 16.7% and leaves the other two rates unchanged. This check identifies a permissive feature in the original validator without changing the ordering of the conditions.

The paper contributes a systems construct that connects public-data governance with task-technology fit, information quality, and digital infrastructure. It also offers a six-dimensional audit instrument and a task-level experimental design that reports completion together with specific failure modes. The evidence supports a practical governance standard

based on executable schemas, stable identifiers, verified joins and reference assets, explicit uncertainty, and testable examples. For public managers, quality should be assessed at the level of the task, data, agent, and validation arrangement.

2. Theory of Open Data and Agent Infrastructure Fit

2.1 Open data as a system of effective use

The open-data literature has consistently resisted the equation of availability with value. Gurstein (2011) framed the normative problem as effective use: disclosure can empower already capable users while leaving others unable to convert data into action. Reviews identify recurring barriers in discovery, documentation, interoperability, skills, institutional incentives, and feedback between publishers and users (Attard et al., 2015; Safarov et al., 2017; Zuiderwijk et al., 2015). Ecosystem studies further show that open-data value is co-produced by agencies, portal operators, intermediaries, developers, communities, and complementary services (Bonina & Eaton, 2020; Dawes et al., 2016; Jetzek et al., 2014; Lane et al., 2024).

AI agents change what a portal must supply. Human visitors can read narrative documentation, move through catalog pages, infer abbreviations, and notice contextual warnings. Agents depend more heavily on explicit representations such as resolvable resource IDs, typed fields, query syntax, join constraints, code lists, provenance, and error messages. Public-data infrastructure must support those machine-mediated actions while preserving human accountability.

Information-quality theory defines quality as fitness for use rather than accuracy in isolation (Wang & Strong, 1996). The same dataset may be adequate for browsing, unsuitable for a geospatial join, and risky for an automated policy memo. Agent readability is conditional on the data, the task, and the agent.

Research on artificial intelligence in government identifies a broad application landscape but also recurring problems of accountability, data quality, institutional capacity, and trust (Janssen et al., 2020; Wirtz et al., 2019; Zuiderwijk et al., 2021). Those studies explain why responsible adoption depends on governance around the model. The present study examines a narrower missing mechanism: whether the installed public-data infrastructure supplies the executable objects that an analytical agent needs to complete a civic task without inventing fields, targets, or relationships.

Public-data ecosystems can also be understood as complex adaptive systems. Multiple agencies, platform operators, data stewards, developers, citizens, and machine agents act on local information, respond to feedback, and alter the conditions faced by others (Holland, 1992; Ostrom, 2010). A local documentation choice can therefore have effects beyond a single dataset. Agent-infrastructure fit provides a task-level mechanism for linking those local affordances to completion, error, and governance feedback across the wider system.

2.2 Task-technology fit, affordances, and the expanded system boundary

Task-technology-fit theory holds that performance depends on the match between task requirements and technology functionality (Goodhue & Thompson, 1995). Agent-infrastructure fit retains that logic and expands the analytical boundary. The technology includes the model and the public-data infrastructure on which it acts. The task includes discovery, selection, query formation, interpretation, joining, validation, and communication. The user is a human-agent arrangement in which the agent proposes operations and a person or automated validator accepts, rejects, or revises them.

Affordance theory helps specify the mechanism. Technical objects offer possibilities and constraints that become consequential in use (Markus & Silver, 2008). In civic data, an endpoint permits retrieval, a typed schema supports query construction, a stable identifier

supports reproducibility, and a code table supports interpretation. Missing or ambiguous objects invite the agent to guess, which creates a governance risk.

Research on digital infrastructure explains how shared systems evolve through layered components, installed bases, standards, and complementary innovations (Henfridsson & Bygstad, 2013; Tilson et al., 2010; Yoo et al., 2010). Public APIs permit outside actors to recombine government data for uses that publishers did not anticipate (Zittrain, 2006). Model Context Protocol tools and similar wrappers translate external data and tools into interfaces that agents can call (Anthropic, 2024; Model Context Protocol, 2026). A wrapper cannot repair a missing schema or unstable identifier. Its performance depends on the underlying infrastructure.

Agent-infrastructure fit has three nested levels. At the asset level, the dataset must expose fields, identifiers, semantics, and operating status. At the task level, those features must support the operations needed to answer a question. At the governance level, the portal must maintain its assets, document changes, report uncertainty, and accept feedback. Strength at one level does not correct a binding failure at another.

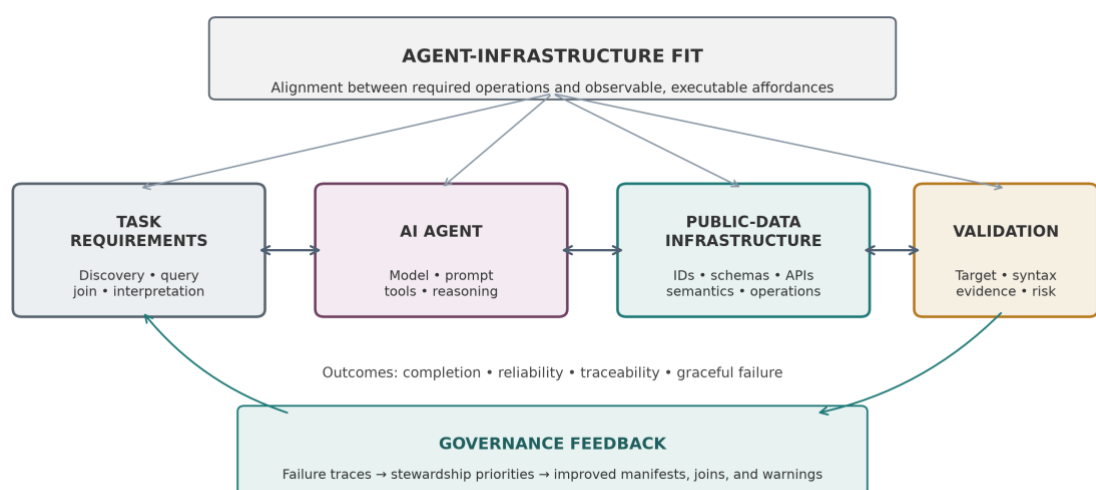


Figure 1
Agent-Infrastructure Fit as a Governed Task System

2.3 Six dimensions and a performance-reliability frontier

We operationalize agent-readable infrastructure through six dimensions. Metadata completeness supports discovery and basic interpretation. Schema documentation supports field selection and type-aware query construction. API queryability supports executable retrieval. Joinability supports relational work across agencies and geographic units. Semantic/reference support supplies code dictionaries, crosswalks, and boundaries. Operational readiness captures recency, non-empty data, update processes, and stability. Table 1 maps each dimension to its mechanism and observable failure mode.

Table 1
Dimensions of Agent-Readable Infrastructure

Dimension	Observable signals	Agent affordance	Failure if weak
Metadata completeness	Description, category, owner, cadence, tags	Discovery and basic interpretation	Wrong asset; missing context
Schema documentation	Typed fields, field descriptions, scale	Field selection and valid query construction	Invented or mistyped fields
API queryability	Stable endpoint, live response, examples	Executable retrieval	Invalid endpoint or syntax
Joinability	Stable keys, types, cardinality, crosswalks	Relational analysis	Wrong or many-to-many join
Semantic/reference support	Code lists, lookups, boundaries, definitions	Interpretation of encoded and spatial values	Missing or misread reference asset
Operational readiness	Recency, non-empty data, update and deprecation signals	Reliable production use	Stale, empty, or unstable result

Note. ARI is an exploratory equal-weight index. The dimensions are complements; a strong score on one does not compensate mechanically for a binding failure on another.

The dimensions are complementary. A stable endpoint with weak semantic support can return valid but uninterpretable codes. Rich documentation paired with an outdated schema can produce plausible syntax and invalid fields. A concise card can prevent invented fields while omitting a reference asset needed for completion. Information designs that improve immediate completion may still permit serious errors, while stricter designs may withhold information needed to finish the task. We describe this tradeoff as a performance-reliability frontier.

The paper tests three propositions.

Proposition 1. Agent-readable affordances improve protocol-defined task completion relative to minimal context, but their effects vary by task family.

Proposition 2. Schema exposure and structured semantic constraints occupy different positions on a performance-reliability frontier: schemas support completion and reference discovery, while compact constraints reduce hallucinated fields and wrong joins.

Proposition 3. Static readiness measures and aggregate portal traffic are insufficient validation targets; a credible governance assessment requires performative, task-level measures of completion and failure.

The third proposition follows from multimethod systems measurement. Frantz et al. (2026) find that self-reported and performative measures of systems thinking are related but distinct. The same distinction applies here. ARI describes observable features of the installed base. The benchmark records what one model does with those features under controlled conditions.

3. Materials and Methods

3.1 Evidence architecture and inferential roles

The three evidence sources answer different questions. The controlled benchmark asks whether information conditions change the model's behavior on the same tasks. ARI describes observable properties of agent-readable public infrastructure. Portal analytics examine access patterns during the 2025 diffusion of agentic coding tools. Table 2 states what each source can and cannot establish.

Table 2*Evidence Architecture and Inferential Boundaries*

Layer	Unit and coverage	Question	Inferential role
Controlled benchmark	90 tasks $\times$ 3 conditions $\times$ 3 repeats; 810 runs	How does supplied infrastructure information change behavior?	Primary within-task experimental evidence; automatic structural outcomes
ARI	Six metadata, schema, API, relational, semantic, and operational dimensions	What features constitute agent-readable infrastructure?	Exploratory measurement and task stratification
Portal analytics	Three core city portals; monthly asset-level access modes	Did access composition change around 2025?	Secondary associational context; not user identification or causal AI adoption
Public traces	Release records, repository events, Data.gov/Open GSA interfaces	Are agent-facing public-data interfaces technically plausible and diffusing?	Chronology, mechanism, and external validity only

Note. No layer is used to make an inference outside its stated boundary.

The benchmark is the primary design because it directly observes model outputs and failure modes. ARI provides a measure of infrastructure features and a basis for task stratification. The portal records supply context because they cannot identify users or isolate an AI shock. Public release records for agentic systems (Anthropic, 2025; GitHub, 2025), repository events (GH Archive, 2026), and public-data interfaces such as Data Commons MCP and Open GSA (Shah, 2025; U.S. General Services Administration, 2026) establish chronology and technical plausibility. They do not measure adoption effects.

3.2 Data sources and preprocessing

The portal archive contains 2,520,115 raw monthly metric records from six Socrata/Tyler systems: New York City, San Francisco, Seattle, Connecticut, Austin, and Oregon. The core descriptive model uses New York City, San Francisco, and Seattle; Connecticut, Austin, and Oregon support task construction and appendix checks. Coverage begins between December 2019 and December 2021 depending on portal and runs through January or June 2026. The three core portals contain metadata for 2,391, 568, and 155 unique assets, respectively.

The processed panel is organized by city, asset, month, and access class for API reads, downloads, and page views. The audit found repeated metadata records for 102 city-asset

keys, including nine exact duplicate San Francisco keys, and 27,947 repeated panel rows. We deduplicated the metadata by city and immutable asset ID, then retained one observation for each city-asset-month-access-class key before recomputing the portal models. The clean panel contains 2,187,934 unique rows. The correction changes the main high-friction estimate from 0.176 to 0.179 without changing its interpretation. It changes the ARI traffic coefficient from 0.009 to -0.018, which remains statistically indistinguishable from zero.

Socrata's Asset Access documentation defines the usage measures and notes that anonymous interactions can include users, bots, crawlers, and other unauthenticated entities (Socrata, 2024). We checked endpoint behavior against the platform's public API documentation (Socrata, 2026). We use “access” instead of “user demand” and “technical access” instead of “agent use.” API reads and downloads are technical modes. Page views provide a comparison mode, not a purely human control.

3.3 Agent-Readable Infrastructure Index

ARI is constructed from the metadata snapshot at the city-asset level. Metadata completeness is the share of available descriptive fields populated among description, category, update frequency, field text, tags, owner, and last-update date. Schema documentation density is the log of field-description words per column, augmented by column scale. Queryability combines pre-period API share with evidence of automated publishing. Joinability combines explicit join-key, geospatial, connective-key, and boundary indicators. Semantic/reference support captures code-table, lookup, boundary, reference, and “how to use” signals. Operational readiness combines standardized update-frequency information with a non-empty-row indicator.

Each component is standardized, and ARI is the equal-weight mean of the six standardized component scores. Low, middle, and high groups are city-specific terciles, avoiding a mechanical advantage for portals with richer catalog conventions. This is an

exploratory index, not a certified standard. Several indicators are text-derived, prior API share is partly behavioral, and current metadata is used to characterize historical traffic. These choices make ARI suitable for task stratification and diagnostic use but not for strong causal interpretation.

3.4 Civic Data Agent Benchmark

The benchmark contains 90 tasks drawn equally from San Francisco, New York City, Seattle, Connecticut, and Data.gov. It includes 20 schema lookups, 20 valid API queries, 20 reference or code-table tasks, 15 geospatial or boundary joins, and 15 substantive civic questions. The Socrata tasks cover all three ARI groups. The full set contains 22 low-ARI, 41 middle-ARI, and 27 high-ARI tasks. Data.gov extends the task set beyond municipal portals through catalog records, download metrics, outbound-link metrics, and DCAT-US metadata (Data.gov, 2026a, 2026b, 2026c).

Each task was executed three times in each of three information conditions on June 28, 2026, using the OpenAI Responses API (OpenAI, 2026) and the gpt-5.5 model with low reasoning effort. The run schedule interleaved conditions in a fixed order; it was not randomized. No external tools were enabled in the three primary conditions. The system prompt required a compact JSON response with answer, dataset ID, endpoint, query, fields used, join keys, reference assets, calls made, evidence, and uncertainty.

The log label `no_metadata` is described here as minimal context because the model received the source and a named task but no target dataset identifier, endpoint, or schema. In the schema condition, it received the dataset name, ID, endpoint, complete recorded field list, and goal. In the structured-ARI condition, it received the name, ID, endpoint, ARI score and tercile, asset family, up to 15 allowed fields, inferred join fields, reference cues, warnings, and goal. The structured card used more input tokens on average than schema (257 versus 187) because it added typed cues and warnings even while truncating the field list.

All 810 planned runs completed. The retained raw JSONL contains one unique record per run ID, and the task-condition balance file confirms completion of all planned cells. The primary outcomes are rule-based and automatic. Schema lookup passes require at least two valid fields and no invalid field. API-query passes require a successful live endpoint probe and no invalid field. Reference and geospatial tasks require a machine-recognizable reference asset or join key without an invalid join. Substantive-answer passes require a successful target-endpoint probe, no invalid field, and non-empty evidence or answer text. Standard errors in the benchmark regressions are clustered by the 90 tasks and include source and task-family fixed effects.

The audit found that the original endpoint validator probed the known target asset for API and substantive tasks even when the response did not identify that asset. We retain the original result for comparison and call it the protocol-defined automatic pass rate. A stricter sensitivity score also requires the returned dataset ID or endpoint URL to match the target for schema, API, and substantive tasks. The original rule remains appropriate for reference and geospatial tasks because the correct supporting asset may differ from the focal dataset.

4. Results

4.1 Information condition changes completion and reliability

Table 3 reports the central benchmark. The schema condition has the highest protocol-defined automatic pass rate at 62.2%. Structured ARI cards pass 52.2%, and minimal context passes 43.0%. In a linear probability model with source and task-family fixed effects and task-clustered standard errors, schema raises the pass probability by 19.3 percentage points relative to minimal context (SE = 4.5 points, $p < .001$). Structured ARI raises it by 9.3 points (SE = 4.4 points, $p = .037$). The direct structured-ARI versus schema

contrast is -10.0 points, showing that the compact card does not dominate a full schema on completion.

Table 3

Benchmark Outcomes by Information Condition

Condition	N	Pass	Strict pass	Halluc. fields	Wrong joins	Ref. asset	Seconds	Tokens
Minimal context	270	43.0%	16.7%	0.56	0.09	34.1%	11.29	747
Full schema	270	62.2%	62.2%	0.57	0.15	76.3%	9.15	726
Structured ARI	270	52.2%	52.2%	0.06	0.02	12.2%	6.46	639

Note. Pass is the original protocol-defined validator outcome reported in the submitted abstract. Strict pass additionally requires correct target identification for schema, API, and substantive tasks. Error columns are mean counts per run; reference asset is the share of runs naming a machine-recognizable asset. Tokens are mean total tokens.

Completion does not capture the full result. Structured ARI produces 0.06 hallucinated fields per run, compared with 0.57 under schema and 0.56 under minimal context. The reduction relative to schema is 89.5%. Wrong joins fall from 0.15 per schema run to 0.02 under structured ARI, a reduction of 87.5%. Schema names a machine-recognizable reference asset in 76.3% of runs, compared with 12.2% for structured ARI and 34.1% for minimal context. The card's warnings and allowed-field constraint reduce unsupported invention. Its truncated field list and general reference cues also limit exploratory retrieval.

These results support the frontier proposed in Proposition 2. Schema performs best on protocol pass rate and reference discovery. The structured card produces fewer hallucinated fields and wrong joins, and it is faster on average at 6.46 seconds compared with 9.15 seconds for schema. It uses more input tokens but produces shorter responses. A useful governance design would combine a compact executable schema with verified reference and join information.

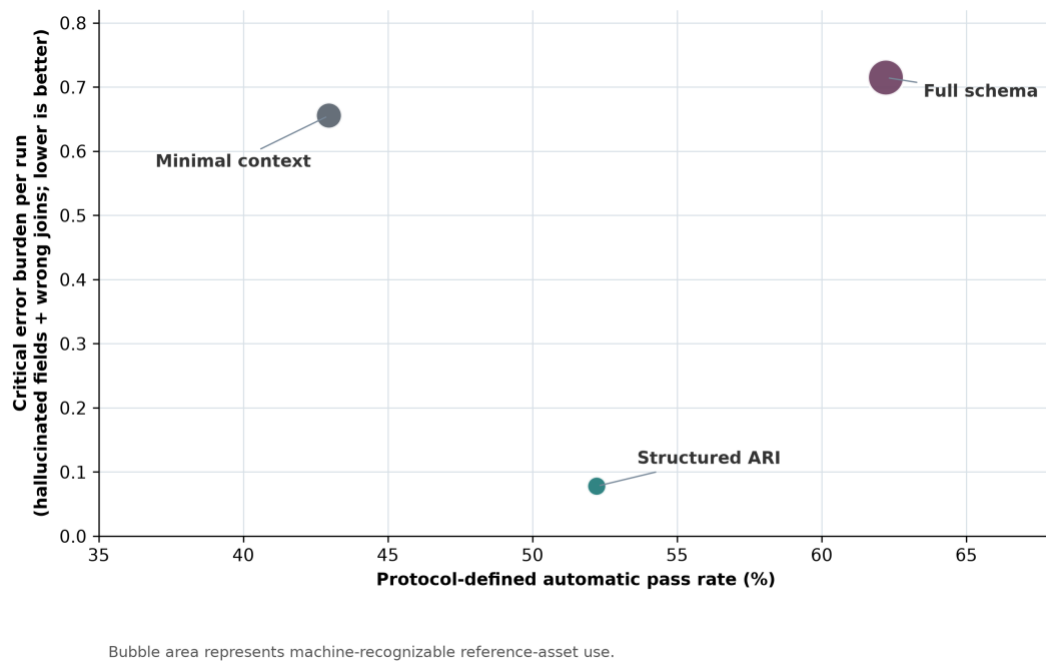


Figure 2
Performance-Reliability Frontier Across Information Conditions

4.2 Task-family heterogeneity

The overall averages conceal sharp differences by task family (Table 4). Schema lookup is impossible under minimal context by the submitted rule and reaches 40.0% under both information-rich conditions. Valid API queries pass at 95.0% under structured ARI and 80.0% under schema, although the minimal-context protocol rate of 91.7% is inflated by the permissive target probe. Substantive-answer tasks pass at 100% under both information-rich conditions by the automatic rule, but this indicates structural compliance, not human-judged semantic correctness.

Table 4
Protocol-Defined Automatic Pass Rate by Task Family

Task family	Runs/condition	Minimal	Schema	Structured ARI
Schema lookup	60	0.0%	40.0%	40.0%
Valid API query	60	91.7%	80.0%	95.0%
Reference/code table	60	25.0%	50.0%	18.3%
Geospatial/boundary join	45	20.0%	46.7%	8.9%
Substantive civic answer	45	82.2%	100.0%	100.0%

Note. Each task is repeated three times per condition. These are structural automatic outcomes, not human-adjudicated semantic accuracy. The minimal-context API and substantive rates are affected by the permissive target probe described in Section 4.3.

Reference and geospatial tasks present the clearest constraint. Schema reaches 50.0% on reference or code-table tasks and 46.7% on geospatial joins. Structured ARI reaches 18.3% and 8.9%, respectively. The treatment design explains the gap. The structured card often names the focal asset or gives a general search instruction, but it does not provide a verified crosswalk, boundary asset ID, or join contract. A readiness label without executable relationship metadata is insufficient for these tasks.

Across all 810 runs, 23 produce an invalid endpoint/API call and the same 23 produce invalid SoQL (2.8% each); 70 contain at least one hallucinated field (8.6%); and 33 contain at least one wrong join key (4.1%). A machine-recognizable reference asset is absent in 479 runs (59.1%). Because many schema and API tasks do not require a reference asset, this last figure is a coverage indicator rather than a universal failure rate. The paper therefore avoids aggregating all failure types into a single severity score.

4.3 Strict identifier sensitivity and validator scope

The strict audit changes only the minimal-context rate. Requiring correct target identification for schema, API, and substantive tasks reduces minimal-context passes from 116 to 45 of 270 runs, or from 43.0% to 16.7%. Schema remains 62.2%, and structured ARI remains 52.2%, because their prompts supplied the correct target ID and endpoint. The condition ordering is unchanged, but the magnitude of the informational advantage becomes larger.

A valid query is useless when it points to the wrong dataset. Endpoint probes must check the target claimed in the response instead of silently substituting the known target. We retain the original protocol rate to avoid changing the reported outcome after inspection and

report the strict rate as a sensitivity analysis. Because no human reviewer graded the substantive civic answers, the study does not estimate semantic answer accuracy.

4.4 ARI predicts when schema is useful, not traffic volume

Table 5 reports the task-level model. ARI by itself does not predict automatic pass in the minimal-context baseline ($b = 0.041$, $SE = 0.058$, $p = .482$). Its interaction with schema is positive ($b = 0.127$, $SE = 0.040$, $p = .002$), while the interaction with structured ARI is smaller and not statistically distinguishable from zero ($b = 0.068$, $SE = 0.049$, $p = .171$). This pattern is consistent with the construct's intended mechanism: observable readiness becomes useful when the agent receives the schema through which those affordances can be acted upon.

Table 5

Task-Level Linear Probability Model of Automatic Pass

Term	b	SE	p	Interpretation
ARI (z)	0.041	0.058	.482	ARI slope in minimal-context baseline
Full schema	0.193	0.045	< .001	Average difference from minimal context
Structured ARI	0.093	0.044	.037	Average difference from minimal context
ARI × full schema	0.127	0.040	.002	Schema advantage rises with ARI
ARI × structured ARI	0.068	0.049	.171	Interaction not distinguishable from zero

Note. N = 810 runs across 90 tasks. Source and task-family fixed effects are included; standard errors are clustered by task. Coefficients are probability-point changes.

The interaction should not be interpreted as independent validation of every ARI component. Tasks are purposively constructed, the model uses a single model family and execution date, and ARI is partly derived from metadata used to select tasks. The result is construct-consistent evidence, not a general causal estimate of infrastructure investment.

4.5 Portal analytics provide context, not causal identification

The deduplicated portal model uses 252,375 observations from 3,108 core city-assets from January 2024 onward, excluding the transition month. The triple interaction compares post-June-2025 change for high- versus lower-friction assets across technical access and

page-view modes, with asset, city-month, and access-class fixed effects and asset-clustered standard errors. The coefficient is 0.179 (SE = 0.033, $p < .001$). The balanced-panel estimate is 0.164 (SE = 0.033, $p < .001$). Mode-specific estimates are 0.244 for API reads, 0.356 for downloads, and 0.151 for page views.

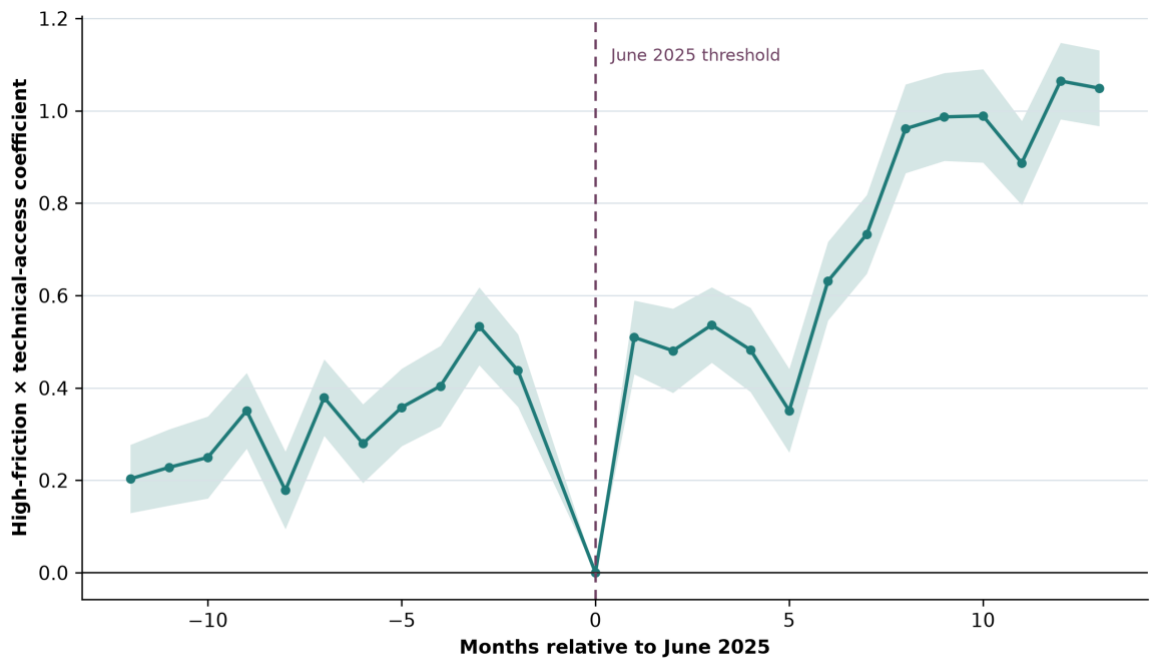
Table 6
Deduplicated Portal-Access Results

Specification	b	SE	p	N / clusters
High friction, main	0.179	0.033	< .001	252,375 / 3,108
High friction, balanced	0.164	0.033	< .001	248,852 / 2,976
Machine composability, main	-0.059	0.032	.065	252,375 / 3,108
Machine composability, balanced	-0.065	0.032	.041	248,852 / 2,976
ARI, main	-0.018	0.033	.587	252,375 / 3,108
ARI, balanced	-0.014	0.033	.679	248,852 / 2,976

Note. Entries are Post $\times$ Group $\times$ Technical-access coefficients from models with asset, city-month, and access-class fixed effects; asset-clustered standard errors. The transition month is excluded. Estimates are associational because pretrends and placebos are nonzero.

The apparent technical-access differential is concentrated in New York City. City-specific high-friction estimates are 0.187 in New York City ($p < .001$), 0.062 in San Francisco ($p = .347$), and 0.306 in Seattle ($p = .186$). The ARI triple interaction is -0.018 (SE = 0.033, $p = .587$), and its page-view coefficient is comparable with its API and download coefficients. Machine composability is negative in the main clean model (-0.059, $p = .065$) and in the balanced specification (-0.065, $p = .041$). These results do not support a claim that high ARI or composability caused a uniquely machine-mediated traffic shift.

The identification diagnostics rule out a causal reading. The joint pretrend test rejects parallel trends ($p < .001$), and the June 2023 and June 2024 placebo interactions differ from zero. Figure 3 shows that the groups diverged before the selected threshold. High-friction assets, especially in New York City, experienced greater technical-access growth during a period when agentic tools were spreading, but the pattern began earlier and may reflect portal-specific or asset-specific changes.



Shaded band: 95% confidence interval. Nonzero leads reject a clean parallel-trends interpretation (joint $p < .001$).

Figure 3
Event-Study Diagnostic for High-Friction Technical Access

The portal records still help to interpret the governance problem. Aggregate traffic cannot validate agent readability because it mixes user types, legacy scripts, popularity, and automated activity. The null ARI traffic coefficient and the condition-specific benchmark results support Proposition 3. Readiness requires direct task tests and recorded failure modes.

5. Discussion

5.1 Fit as a property of the system

The paper locates performance in the relationship among the task, data infrastructure, agent, and validation regime. Public-sector discussions often treat AI readiness as an organizational capability or a property of the model. Agent-infrastructure fit explains why the same model can work reliably on one dataset and fail on another when the civic questions appear similar.

The construct extends task-technology fit by enlarging the technology boundary and placing governance inside the system. A portal's identifiers, schemas, crosswalks, examples, and update practices shape what an agent can do validly (Markus & Silver, 2008). Its output also records where the infrastructure failed. Repeated invalid fields point to schema problems. Wrong joins point to missing relationship contracts. Omitted reference assets point to weak semantic documentation. Publishers can use those records to set maintenance priorities.

This feedback process adds an important condition to digital-infrastructure generativity. Outside actors can build on an infrastructure, but publishers also need to inspect and correct unexpected uses. AI increases the speed and scale of recombination and makes unresolved ambiguity more costly. Agent readability is a governance property of the installed base.

The complex-adaptive-systems lens clarifies why this property cannot be secured by a one-time documentation exercise. Agents, interfaces, schemas, and stewardship practices change together. Failure traces create feedback that can improve manifests, joins, warnings, and validators, while unobserved failures allow brittle local conventions to propagate. The governance task is to make those feedback loops visible and actionable without confusing aggregate activity with reliable task performance (Holland, 1992; Ostrom, 2010).

5.2 Combining an index with direct task tests

ARI provides a compact vocabulary for auditing the installed base. It can support stratification, portfolio screening, and diagnosis, but it cannot replace task tests. An inventory of infrastructure features and the observed performance of the full system are different measures.

The benchmark records several outcomes: protocol pass, target identification, invalid API use, invalid syntax, hallucinated fields, wrong joins, reference-asset use, time, and

tokens. A single accuracy score would hide the tradeoff among them. The condition with the highest completion rate also produces more hallucinated fields and wrong joins than the compact card. An agency may reasonably accept a lower completion rate when the avoided errors affect benefits, policing, property, or health decisions.

The strict identifier audit shows that the evaluator can also affect results. In this study, the target endpoint probe repaired omissions in minimal-context responses and inflated their reported success. Agent benchmarks should treat validators as part of the study design, state their rules, and report sensitivity checks when those rules can change an outcome.

5.3 Governance design principles

The evidence supports five design principles for public-data leaders.

First, publish an executable schema manifest. It should pair a stable asset ID and endpoint with typed fields, temporal coverage, update status, and one minimal valid query. The benchmark shows that schema access produces the highest completion rate.

Second, publish relationship metadata as first-class infrastructure. A join contract should identify the left and right assets, keys, key type, cardinality, geographic and temporal scope, and known exceptions. Generic cues such as “district” or “parcel” are not enough.

Third, separate descriptive metadata from operational constraints. Human-readable descriptions remain valuable, but agents also need an allow-list of fields, warnings about deprecated or historical assets, and explicit uncertainty. Structured ARI cards sharply reduce hallucinated fields and wrong joins.

Fourth, validate the claimed target and the returned evidence. A validator should never substitute a ground-truth endpoint without recording the repair. Substantive civic answers should require evidence-level checking or human review, especially in high-stakes domains.

Fifth, govern for graceful failure. When a code table, boundary, or join cannot be verified, the desired behavior is a precise abstention with a missing-asset message. Portal teams can aggregate those messages into a maintenance backlog. Table 7 translates the failure taxonomy into operational controls.

Table 7
From Benchmark Failure to Governance Control

Observed failure	Infrastructure diagnosis	Recommended control	Machine-testable acceptance criterion
Wrong or absent target	Discovery and identity are underspecified	Immutable ID; canonical endpoint; aliases and deprecation redirect	Returned ID and host match manifest
Hallucinated field	Schema is absent, stale, or unconstrained	Typed allow-list; schema timestamp; example query	Every referenced field exists at validation time
Invalid endpoint/SoQL	Retrieval contract is incomplete	Minimal valid query; syntax and rate-limit guidance	Example executes and returns typed data
Wrong join	Relationship semantics are implicit	Join contract with key type, cardinality, scope, and exceptions	Sample join passes uniqueness and coverage tests
Missing reference asset	Codes or boundaries are not packaged relationally	Verified lookup/boundary IDs and versioned crosswalks	Reference resolves and covers expected keys
Unsupported substantive answer	Evidence validation is too shallow	Evidence entailment and human review for high-stakes use	Claims map to returned rows and provenance

Note. Controls are framed as tests so that failure traces can be aggregated into a stewardship backlog.

5.4 Relation to AI productivity research

Prior experiments find that generative AI can improve productivity while producing uneven or adverse results on some tasks (Brynjolfsson et al., 2025; Dell'Acqua et al., 2026; Noy & Zhang, 2023; Peng et al., 2023; Vaithilingam et al., 2022). This study identifies one infrastructural source of that variation. Public agencies influence the capability boundary through the data features they publish and maintain.

Completion, semantic discovery, and error control respond differently to the same information design. This systems result has a direct managerial consequence: optimizing one benchmark outcome can move risk elsewhere in the workflow.

6. Limitations and Research Agenda

The benchmark evaluates one model alias at one reasoning setting on one execution date, with three repetitions per task and condition. The results estimate how information design affected this model under the stated protocol. They do not establish that every model would respond in the same way. A replication should compare multiple model families, use frozen versions when available, randomize execution order, and increase the number of repetitions.

The automatic pass rate measures structural compliance. No human reviewer graded the substantive civic answers, so the study does not claim final semantic correctness. Later studies should use blinded expert grading, report agreement among raters, check whether the evidence supports each answer, and distinguish errors by severity. Validator rules should also be registered before execution.

The structured ARI treatment uses one representation. It limits the field list to 15 entries, infers join candidates with regular expressions, and often supplies general rather than verified reference cues. A stronger treatment would provide a minimal complete schema with explicit join contracts and reference-asset IDs. A factorial design could then estimate the separate effects of field-list breadth, semantic constraints, examples, warnings, and reference metadata.

ARI is constructed from current metadata and includes prior API share, making it partly endogenous to existing use. Text-derived components may reflect portal-specific documentation conventions. The index should be validated with confirmatory factor analysis, out-of-sample tasks, temporal snapshots, expert ratings, and predictive tests across non-Socrata infrastructures.

The portal study cannot identify users or causal AI adoption. Nonparallel pretrends, significant placebos, city heterogeneity, and anonymous automated traffic rule out a clean

difference-in-differences interpretation. Future work could use authenticated developer programs, platform-specific feature rollouts, API-key cohorts, or randomized documentation interventions.

The study addresses technical and semantic usability, not the full ethics of civic AI. Making data easier for agents to use may also increase surveillance, reidentification, exclusion, or automated misuse. Any readiness standard should address privacy, equity, security, contestability, and public purpose alongside usability.

7. Conclusion

AI agents depend on the identifiers, schemas, joins, reference assets, operating signals, and validation procedures around them. Agent-infrastructure fit gives public agencies a way to examine that relationship as a governance problem.

The controlled benchmark identifies a tradeoff between completion and error control. Full schemas support completion and reference discovery. Compact structured constraints reduce hallucinated fields and wrong joins but omit relationship information needed for some tasks. The strict audit shows that the evaluator must check the target identified by the model. After deduplication, the portal evidence is consistent with changing technical access but does not support a causal AI claim.

Public managers should test representative tasks, inspect failure modes, publish executable relationship metadata, and use observed failures to guide stewardship. Dataset counts and page views do not establish whether an agent can use public data reliably.

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Appendix A. ARI Operationalization

ARI is computed after deduplicating metadata at the city-asset level. Each final component is standardized before equal-weight averaging.

Table 8

ARI Component Operationalization

Component	Operationalization in retained scripts	Important limitation
Metadata completeness	Share of available descriptive fields populated	Field availability and conventions differ by portal
Schema documentation	Log field-description words per column plus column scale	Volume is not the same as semantic accuracy
API queryability	Pre-period API share plus automated-publishing signal	Partly behavioral and endogenous to prior use
Joinability	Join-key, geographic, connective-key, and boundary indicators	Primarily text- and pattern-derived
Semantic/reference support	Code, lookup, boundary, reference, and how-to-use signals	Presence does not verify a correct relationship
Operational readiness	Update-frequency standardization plus non-empty-row indicator	Current snapshot may not reflect historical state

Note. Each component is standardized and equally weighted. Low, middle, and high labels are within-city terciles.

The tercile labels are ranked within city. The index is exploratory and is not interpreted as a causal treatment. Because prior API share and the current metadata snapshot can reflect past use and later documentation, the traffic regressions use ARI only as a diagnostic.

Appendix B. Benchmark Protocol and Audit Trail

The retained benchmark status identifies version v2_structured_ari, prompt version structured_ari_card_v2, validator version validator_v2, model gpt-5.5, three repeats, 90 tasks, and 810 completed runs. The raw JSONL contains the prompt-condition output, token use, automatic validation, and run metadata for each unique run ID. The audit independently recomputed condition summaries and the task-clustered model; differences from stored coefficients and standard errors were below 5×10^{-16} .

Table 9

Benchmark Validator Rules and Known Boundary

Task family	Original automatic rule	Strict audit change	Interpretation boundary
Schema lookup	≥ 2 valid fields and no invalid field	Also require target ID/endpoint match	Schema structure, not conceptual relevance
Valid API query	Known-target probe succeeds; no invalid field	Require claimed target match before probe credit	Execution validity, not analytical usefulness
Reference/code table	Recognizable reference asset; no invalid join	No change	Naming is not proof of correct use
Geospatial join	Recognizable join/reference; no invalid join	No change	Does not validate spatial coverage
Substantive answer	Known-target probe; no invalid field; nonempty answer/evidence	Require target match	Not human-judged semantic accuracy

Note. The known-target probe in the original validator can repair a missing target implicitly. The strict score exposes that behavior without replacing the reported primary outcome.

The strict sensitivity is post-hoc and deliberately narrow. It does not rescore reference or geospatial tasks because the correct supporting asset may differ from the focal dataset. It is reported to expose the target-substitution behavior of the original validator, not to replace the submitted primary outcome.

Appendix C. Portal Model and Robustness Interpretation

The clean portal model estimates:

$$\begin{aligned} \ln(1 + Access_{idmt}) \\ = \beta(Post_t \times Group_i \times TechnicalAccess_m) + \alpha_i + \lambda_{dt} + \mu_m \\ + \varepsilon_{idmt}. \end{aligned}$$

Group is alternately high technical friction, high machine composability, or high ARI.

Standard errors are clustered by asset. The transition month is excluded. Results are associational because the event-study leads and pre-period placebo thresholds are nonzero.

The paper therefore does not use causal verbs for the portal estimates.

Table 10

Portal Robustness and Diagnostic Estimates

Contrast/specification	b	SE	p	Interpretive status
High friction: API reads	0.244	0.036	< .001	Mode-specific association
High friction: downloads	0.356	0.037	< .001	Mode-specific association
High friction: page views	0.151	0.034	< .001	Comparison-mode association
High friction: San Francisco	0.062	0.066	.347	Not distinguishable from zero
High friction: New York City	0.187	0.035	< .001	Primary city concentration
High friction: Seattle	0.306	0.233	.186	Imprecise
Joint pretrend test	NA	NA	< .001	Rejects parallel trends

Note. NA means not applicable. All models use the deduplicated analysis copy. Significant pre-period coefficients and placebo thresholds preclude a causal AI interpretation.